CAPSTONE PROJECT | MASTER OF ANALYTICS

Optimizing Home Loan Decisions with Machine Learning

Default prediction, risk-based borrower segmentation and regulatory stress testing for the Australian mortgage market

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Auckland University of Technology, in partnership with SAS

With thanks to Dr. Shu Su, Lucy Biasi, Janhav Khanna, Lachlan Catto, Jennifer Chuor



OVERVIEW

Today's agenda

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Introduction & project motivation

02

Model development and key results

03

Borrower segmentation insights

04

Regulatory and economic impact

05

Housing market health & policy actions

06

Conclusion and future directions

Project background & motivation

SAS leadership

- Global leader in advanced analytics and AI solutions
- Trusted by major Australian banks for credit risk and compliance

Market challenges

- Rising interest rates and tighter lending regulations
- Increasing borrower defaults and affordability pressures
- Growing demand for faster, data-driven loan decisions

Banks need lending decisions that are both fast and safe.

Project objectives

OBJECTIVE 1



Develop & evaluate loan origination models

Build and compare default prediction models using SAS Viya and Workbench.

OBJECTIVE 2



Segment borrowers by risk profile

Turn predicted risk into four actionable borrower segments for fairer lending.

OBJECTIVE 3



Analyse regulatory & economic impacts

Stress-test lending rules and economic shocks to guide housing policy.

Dataset overview: SAS HMEQ home equity loans

5,960
loan applications

13
variables

20%
default rate (realistic imbalance)

Target variable

BAD = 1 defaulted | BAD = 0 paid back

Predictor variables

Financial: LOAN, MORTDUE, VALUE, DEBTINC
Employment: JOB, YOJ
Credit: DEROG, DELINQ, CLAGE, NINQ, CLNO
Loan purpose: REASON

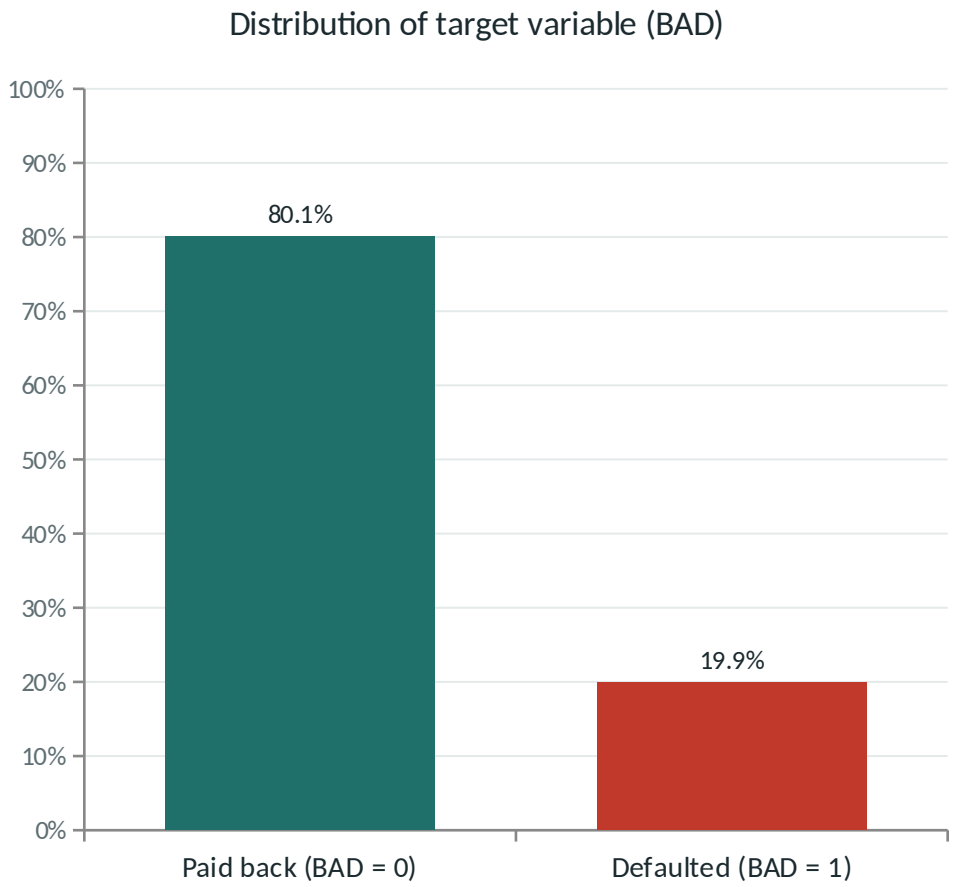


Figure 1: Distribution of target variable (BAD)

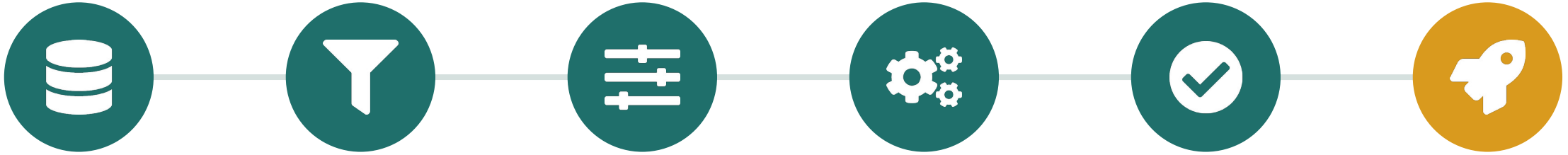
01

Building smarter home loan origination models

Using SAS Viya and Workbench to enhance accuracy and efficiency in loan approvals



Pipeline for loan origination models



STEP 1

Raw data collection

STEP 2

Data cleaning & preparation

Missing value imputation, outlier treatment

STEP 3

Feature engineering & selection

WOE binning, IV ranking

STEP 4

Model training & optimization

4 candidate models

STEP 5

Validation & performance testing

AUC, Gini, KS, F1

STEP 6

Business integration & deployment

SAS Viya decisioning

Data cleaning: missing value imputation

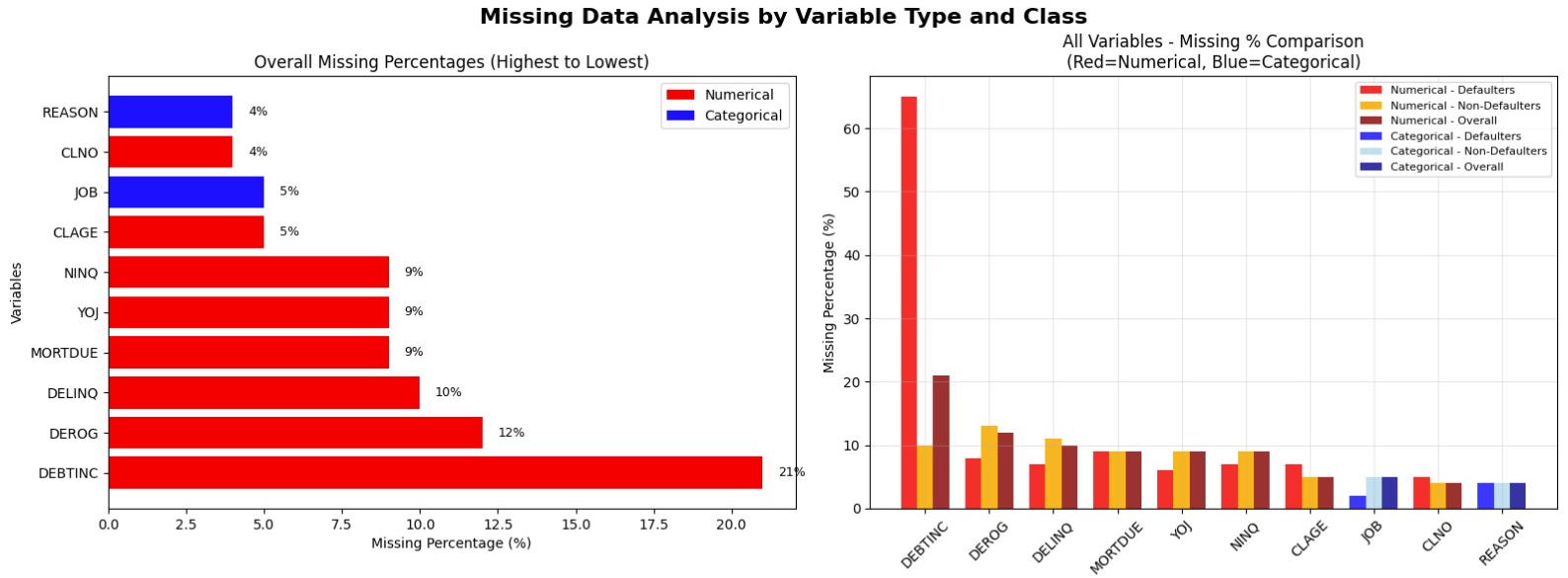


Figure 2: (L) Group-wise missing values (%) by variable | (R) DEBTINC before and after imputation

65% vs 10%

DEBTINC missing for defaulters vs non-defaulters. Missingness itself is a strong risk signal.

Advanced regression-based group-wise imputation

Missing-value indicators created as predictors

Imputation validated: distributions preserved

Outlier detection & treatment

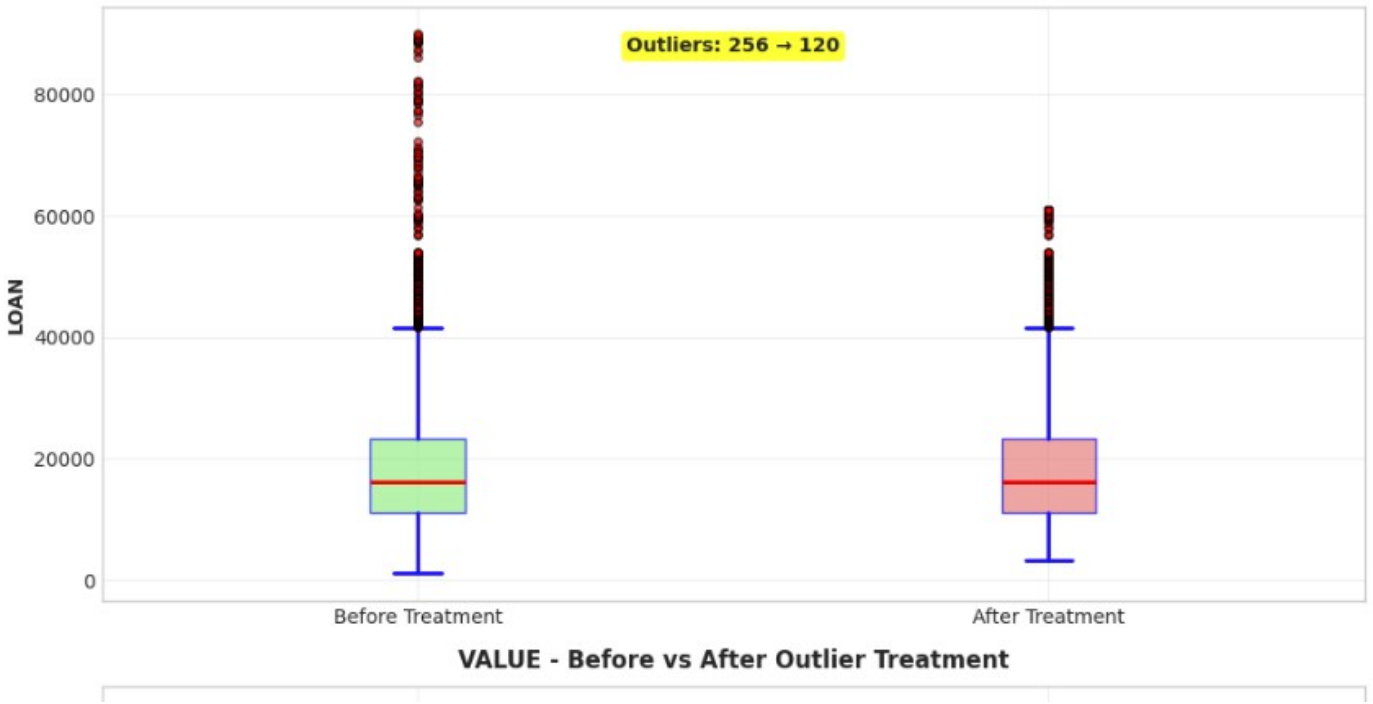


Figure 3: Outlier treatment for VALUE, capping at the 1st and 99th percentiles

Detection

- Outliers found in LOAN, VALUE, DEBTINC, CLAGE
- 47 to 60 extreme cases per variable (1st / 99th percentile)

Treatment

- Winsorized at the 1st / 99th percentile
- All samples kept, only extremes capped
- Risk outliers (e.g. delinquencies) retained as genuine behaviour

Feature engineering & transformation

Binning & discretization

- Weight of Evidence (WOE) binning for continuous variables
- Risk-based categories via quantile binning
- Manual binning for sparse counts (DEROG, DELINQ, NINQ)

Derived variables

- Loan-to-value (LTV) ratio
- Missing-value indicator flags

Benefits

- Improved model interpretability
- Enhanced predictive power
- Better handling of non-linear relationships
- Scorecard-ready features

Weight of Evidence (WOE) & Information Value (IV)

WEIGHT OF EVIDENCE

$$WOE = \ln (\% \text{ of goods} / \% \text{ of bads})$$

Measures how well each bin separates good from bad borrowers. % of goods (bads) = share of all good (bad) customers that fall in the bin.

INFORMATION VALUE

$$IV = \Sigma (\% \text{ of goods} - \% \text{ of bads}) \times WOE$$

Summarises a variable's overall power to predict default, used to rank and select features.

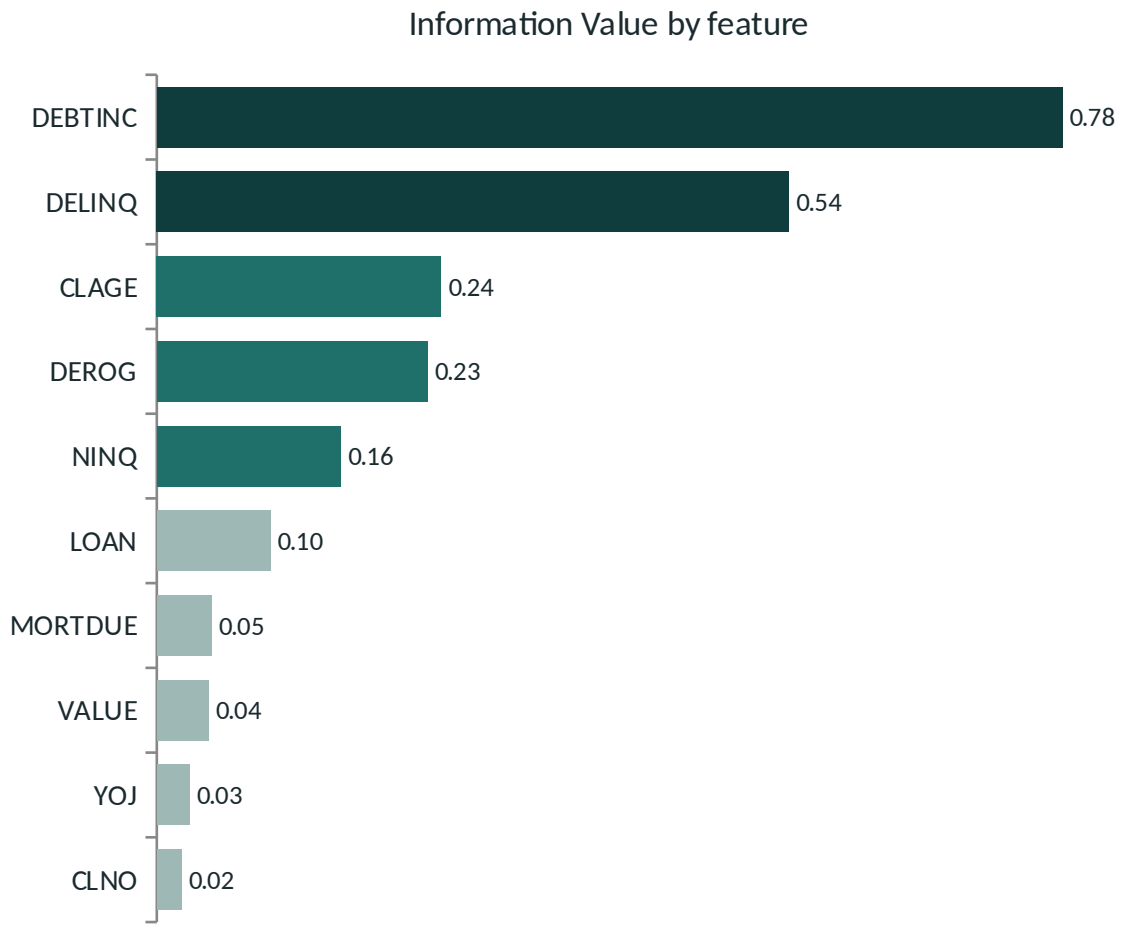
Rules for interpreting IV

Information value	Predictiveness
< 0.02	Not useful for prediction
0.02 to 0.1	Weak predictive power
0.1 to 0.3	Medium predictive power
0.3 to 0.5	Strong predictive power
> 0.5	Suspiciously strong (check for leakage)

Debt burden and credit history are the strongest predictors

Feature	IV	Strength
DEBTINC	0.7751	Very strong
DELINQ	0.5411	Very strong
CLAGE	0.2434	Medium
DEROG	0.2317	Medium
NINQ	0.1575	Medium
LOAN	0.0972	Weak
MORTDUE	0.0469	Weak
VALUE	0.0443	Weak
YOJ	0.0281	Weak
CLNO	0.0211	Weak

Table 1: IV values and feature strength in predicting loan default



Model development overview

Data split

4,172
training samples (70%)

1,788
test samples (30%)

- Stratified sampling keeps a 20% default rate in both sets
- Class weighting handles the imbalance

Logistic regression
Baseline scorecard model

Decision tree
Interpretable splits

Random forest
Ensemble method for accuracy

Neural network / gradient boosting
Strong predictive power

Random forest is the champion model

Model	Accuracy	F1-score	ROC-AUC
Logistic regression	85.1%	69.3%	91.0%
Decision tree	91.1%	78.0%	87.8%
Neural network	94.9%	86.1%	95.7%
Random forest	**95.0%**	**86.6%**	**96.6%**

Table 2: Test-set performance across candidate models

Highest ROC-AUC and F1 of all models, balancing accuracy, stability and business performance.



CHAMPION MODEL

Random forest

0.966
ROC-AUC

0.866
F1-score

Champion model evaluation: random forest

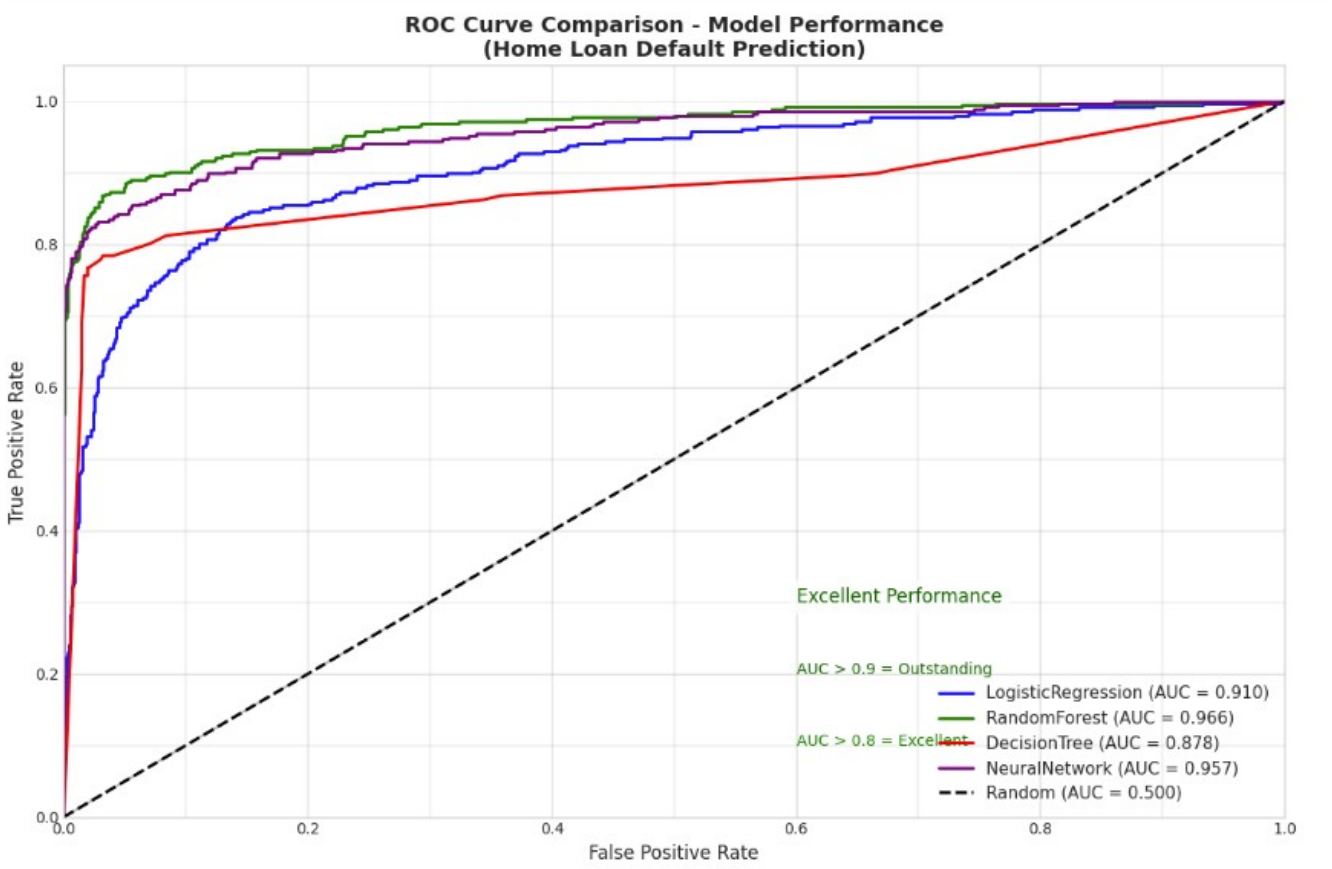


Figure 4: ROC curve comparison across models

0.966
ROC-AUC
Excellent classification

0.931
Gini
Outstanding discrimination

0.837
KS statistic
Strong separation

95%
Accuracy
Overall

80%
Recall
Bad loans caught

0.994 / 0.966
Train / test AUC
Low overfitting

Stable, accurate and ready for deployment in SAS Viya

What Objective 1 delivered

Higher predictive accuracy

Advanced models such as neural network and random forest outperformed the logistic baseline.

Excellent risk discrimination

Random forest delivered the best separation of good and bad borrowers with strong stability.

Fairer, more accurate approvals

Better-calibrated risk reduces both wrongful rejections and risky approvals.

Data-driven origination

Supports an efficient home loan origination process in the Australian market.

02

A smarter lending strategy: risk-based borrower segmentation

Using machine learning to drive profitability and manage risk



From prediction to action



Leverage

our high-performance default prediction model



Segment

borrowers into distinct risk profiles



Enable

targeted product offers, pricing and automated decisions

Turning predictive scores into actionable lending strategies.

Borrower segmentation using K-means clustering

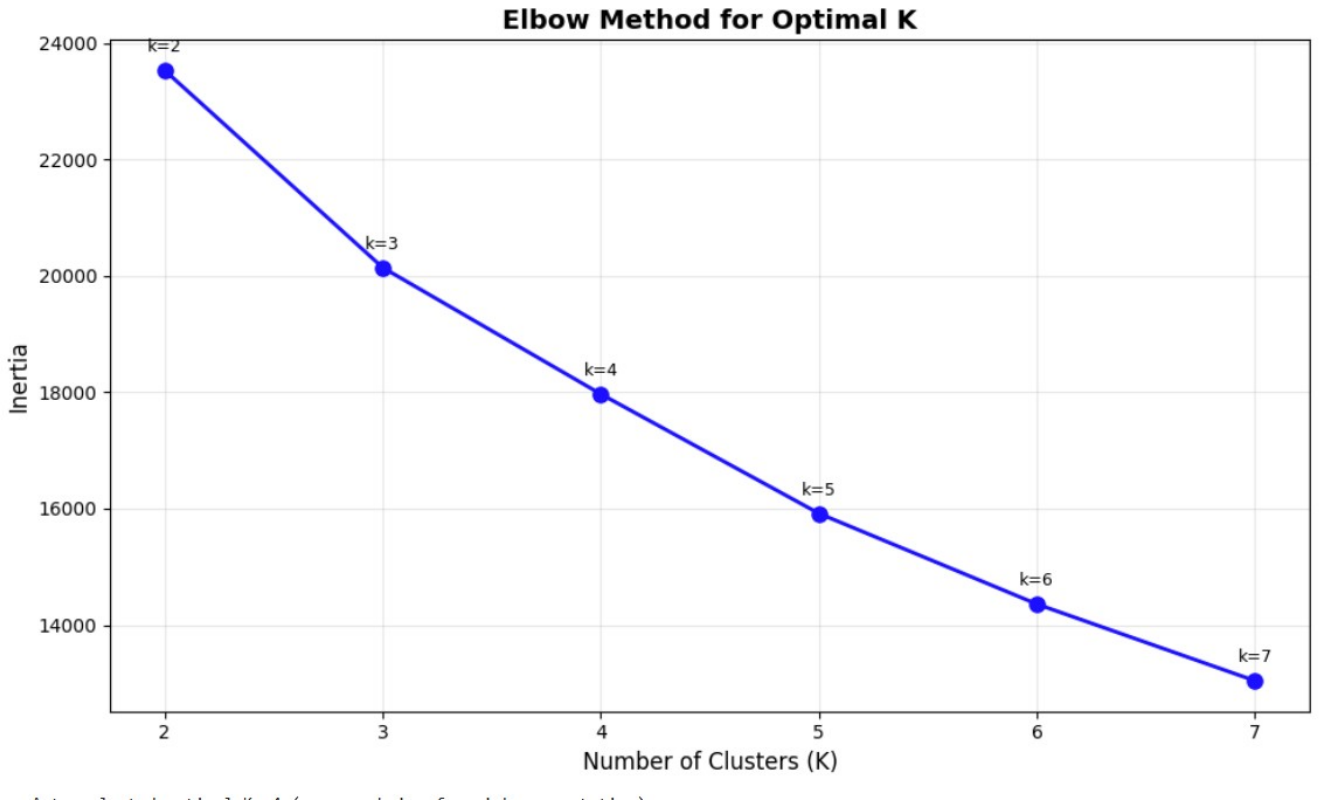


Figure 5: Elbow method for the optimal number of clusters (K)

Elbow method

Clear bend at K = 4

Silhouette score

0.19 at K = 4 (acceptable)

Davies-Bouldin / Calinski-Harabasz

1.617 / 868.6, well-separated segments

Business fit

Industry standard of 4 actionable risk tiers

Introducing the four borrower personas

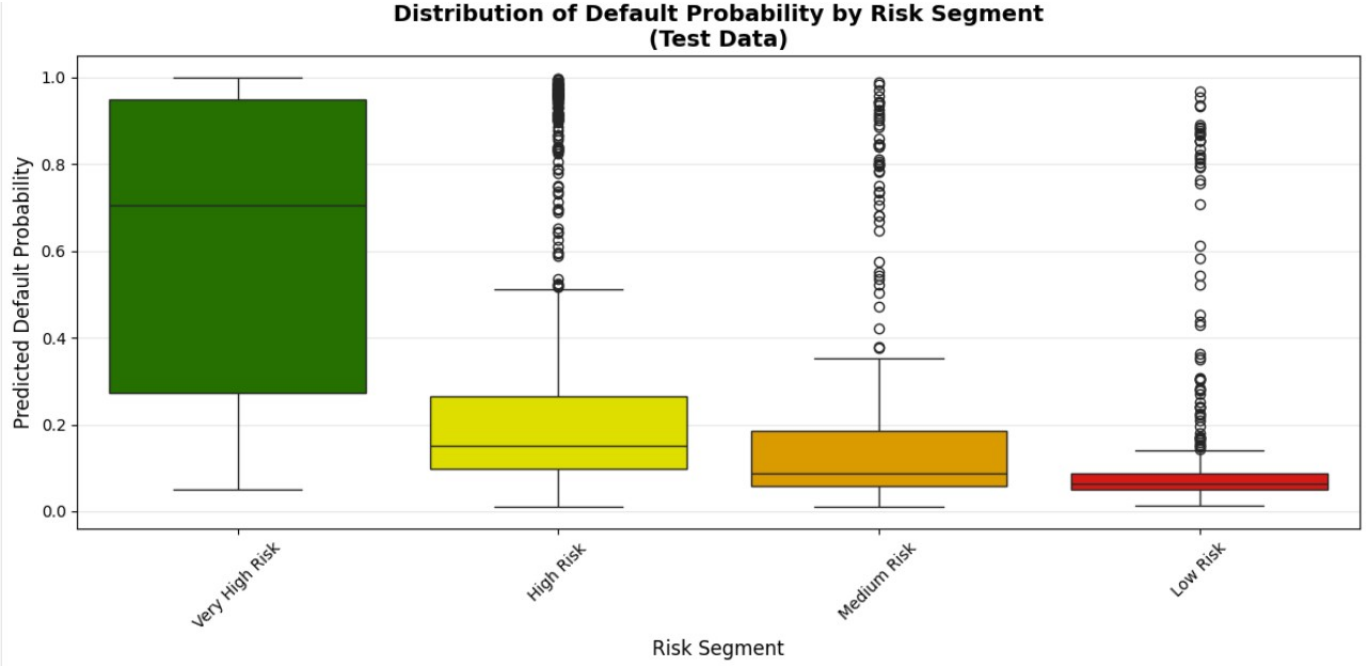


Figure 6: Predicted default probability distribution by risk segment

Default probability rises steadily from Secure to Vulnerable borrowers, giving each persona a clear lending treatment.

LOW RISK

The Secure Borrower

Stable finances, clean history

MEDIUM RISK

The Managing Borrower

Moderate debt, some credit hiccups

HIGH RISK

The Strained Borrower

High debt burden, notable credit issues

VERY HIGH RISK

The Vulnerable Borrower

Severe financial distress

Segments match real-world default outcomes

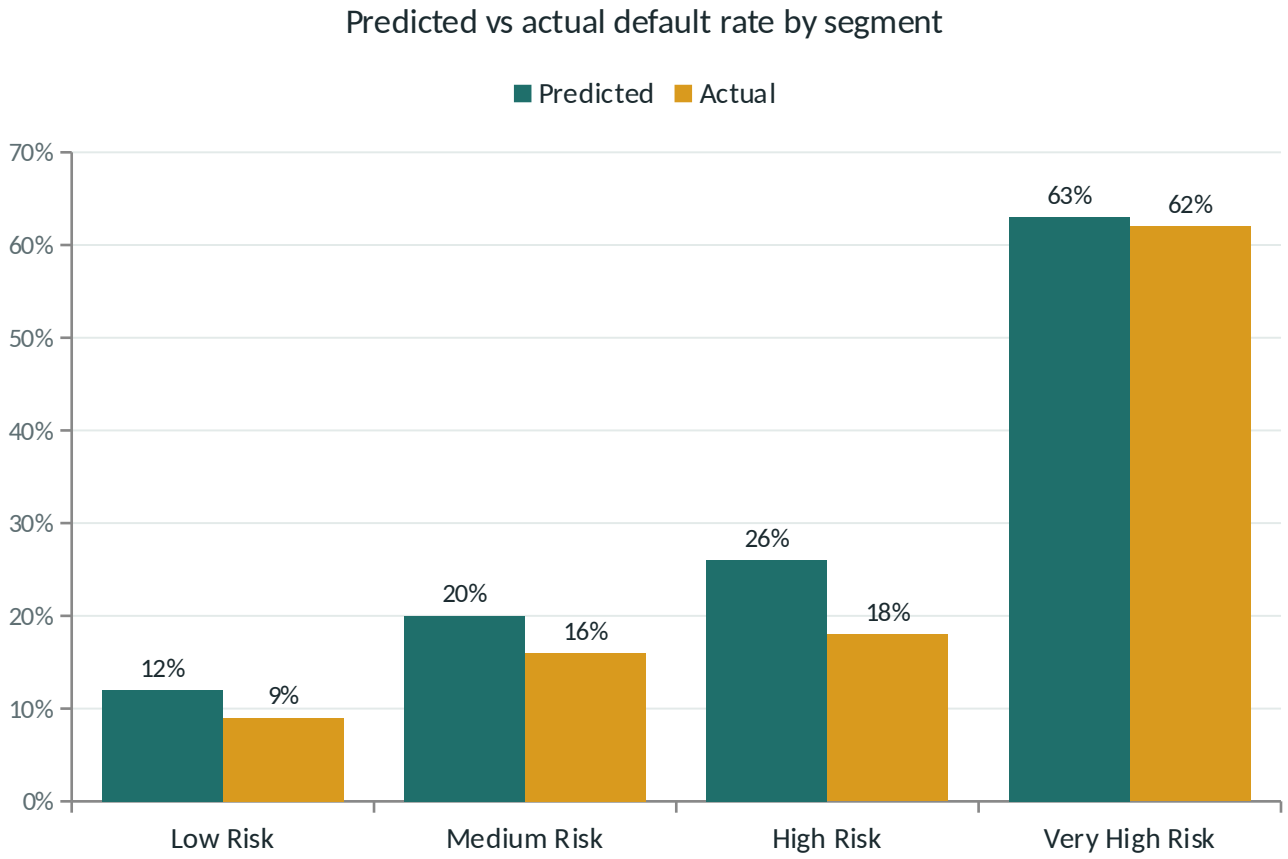


Figure 7: Predicted vs actual default rates (test set)

Low Risk	452 applicants	**+3 pts**
Medium Risk	299 applicants	**+4 pts**
High Risk	844 applicants	**+8 pts**
Very High Risk	193 applicants	**+1 pt**

MAE 0.040

Average error of about 4 points; highest-risk segment predicted almost exactly.

Strategic decision framework

1. Segment by risk profile
2. Define action
3. Assign terms
4. Monitor & adjust

Risk segment	Portfolio	Lending action	Pricing	Monitoring
Low Risk	29.3%	Auto-approve: instant decision	6.5 to 7.5%	Light touch, quarterly review
Medium Risk	42.4%	Approve with conditions: extra verification	7.5 to 9.0%	Standard monitoring, payment tracking
High Risk	18.1%	Manual review: senior assessment	9.0 to 11.5%	Active monitoring, early-warning alerts
Very High Risk	10.2%	Decline: protect portfolio	12.5%+ if approved	Intensive monitoring, default prevention

Business impact summary

67.3%
of customers get instant decisions

20-25%
reduction in portfolio default risk

29.1%
margin improvement on approved loans

55.1%
better capital utilization

Metric	Before	After	Improvement
Default rate	20.0%	13.7%	**20-25% reduction**
Profit margin	7.0% avg	9.04% avg	**29.1% increase**
Auto-approval	Manual process	67.3% automated	**Efficient process**
Capital efficiency	$50.3M RWA	$22.6M RWA	**55.1% better capital use**

Table 3: Business benefits. RWA = risk-weighted assets; APRA = Australian Prudential Regulation Authority. Calculations in the appendix.

What Objective 2 delivered



Four risk segments

Low, Medium, High and Very High risk, each with a clear persona.



Validated against reality

Segments match actual defaults with an average error of about 4%.



Tailored actions

Fast-track, conditional, senior review or minimise exposure.



Portfolio benefits

Faster decisions, fewer defaults, higher margins, efficient capital.



Fair & compliant

Supports fair, data-driven and compliant lending decisions.

03

Regulatory & economic impact on home loans

1. Assess how lending regulations affect loan origination and affordability
2. Analyse the effect of economic shocks on loan volumes and default risk
3. Provide insights for policy that supports a stable housing market

Four key areas of study



Regulatory impact

How LVR, buffers, DTI limits and stress rates change loan origination

Economic scenarios

Rate shocks, housing corrections, recovery and stagflation

Market health assessment

Scorecard across five dimensions of housing market health

Policy actions

Prioritised recommendations and their cumulative impact

Regulatory impact analysis: three scenarios

Scenario	LVR limit	Serviceability buffer	DTI limit	Stress test rate
Current APRA	0.80	0.03	None	0.07
Tightened regulation	0.70	0.04	6.0	0.08
First home buyer support	0.90	0.025	7.0	0.065

Table 4: Regulatory scenarios on lending standards and stress testing

Current APRA

Baseline regulatory settings: up to 80% LVR with a 3% serviceability buffer.

Tightened regulation

Stricter risk limits: 70% LVR, higher buffer and stress rate, DTI cap of 6.

First home buyer support

Relaxed entry conditions: 90% LVR and lower stress test rate.

Tighter rules improve stability but reduce access

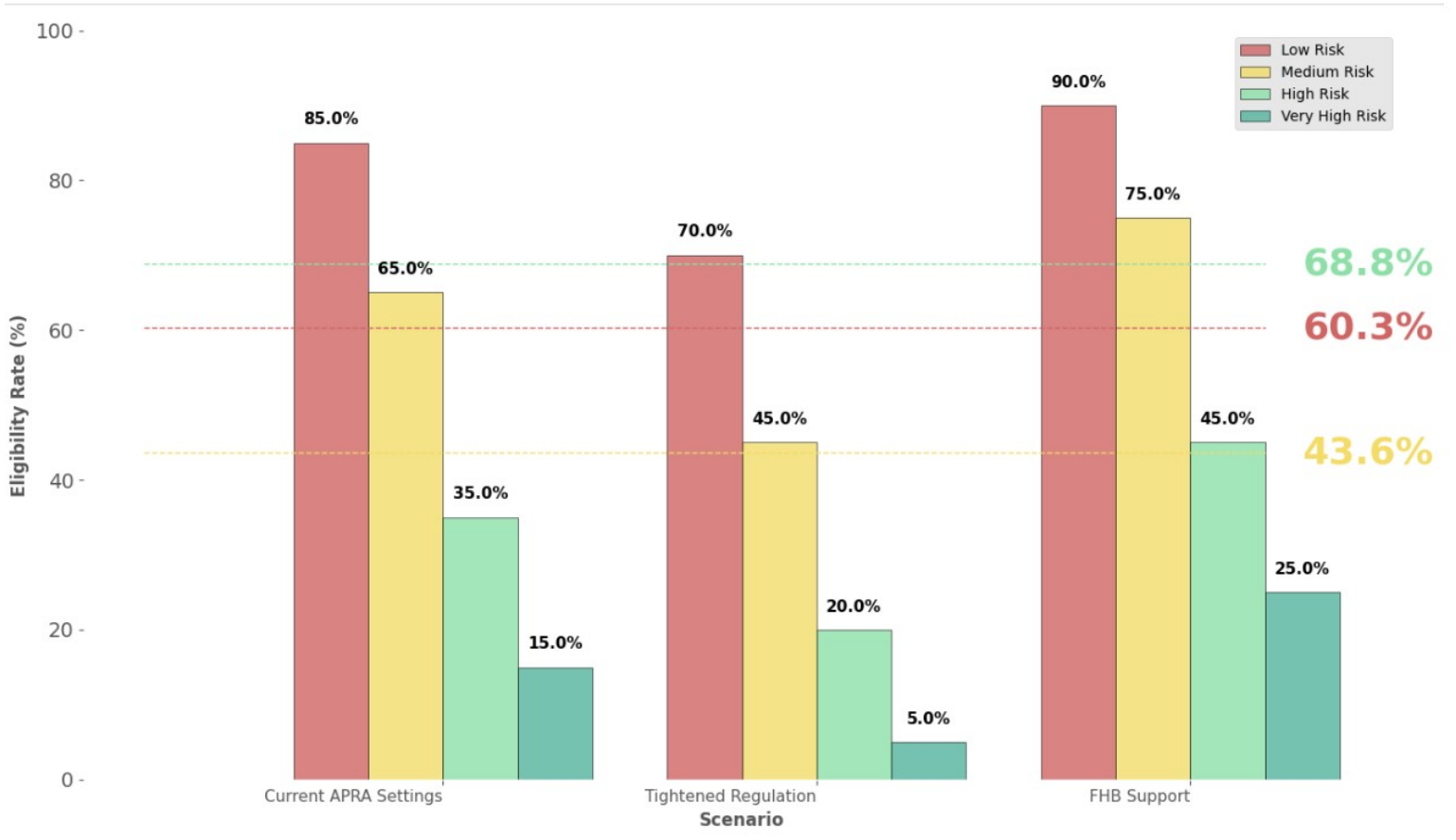


Figure 8: Regulatory impact on borrower eligibility by risk segment

Average borrower eligibility

69% First home buyer support

60% Current APRA

44% Tightened regulation

Low-risk borrowers are barely affected; high-risk borrowers are highly sensitive, so targeted support works better than blanket changes.

Macroeconomic stress scenarios

Scenario	Interest rate	Property prices	Unemployment	Consumer confidence
Interest rate shock	+2.0%	-5.0%	+1.0%	-10%
Housing market correction	+1.0%	-15.0%	+2.0%	-20%
Economic recovery	-1.0%	+8.0%	-1.5%	+15%
Stagflation	+3.0%	-10.0%	+2.5%	-25%

Table 5: Economic scenarios with changes in interest rates, property prices, unemployment and consumer confidence. Green = favourable, red = adverse.



Scenarios are severe but plausible stress tests to assess portfolio resilience, not forecasts of future events.

Economic shocks cut lending and raise default risk

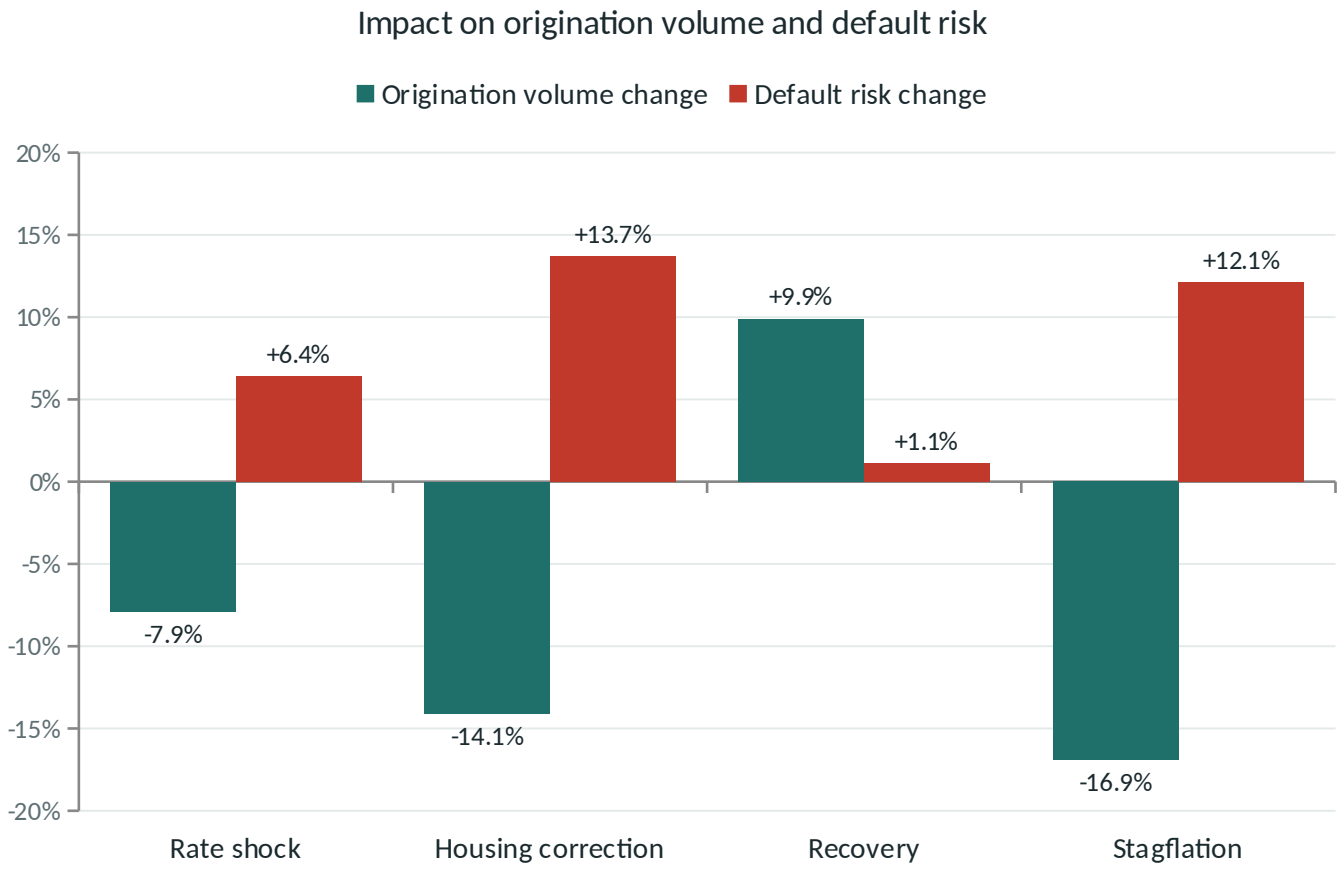


Figure 9: Impact of economic changes on lending and loan risk

+2% interest rate

Origination -7.9%, defaults +6.4%

-15% property values

Origination -14.1%, defaults +13.7%

Economic recovery

Origination +9.9%, defaults +1.1%

Stagflation

Origination -16.9%, defaults +12.1%

Riskier segments are most sensitive to shocks

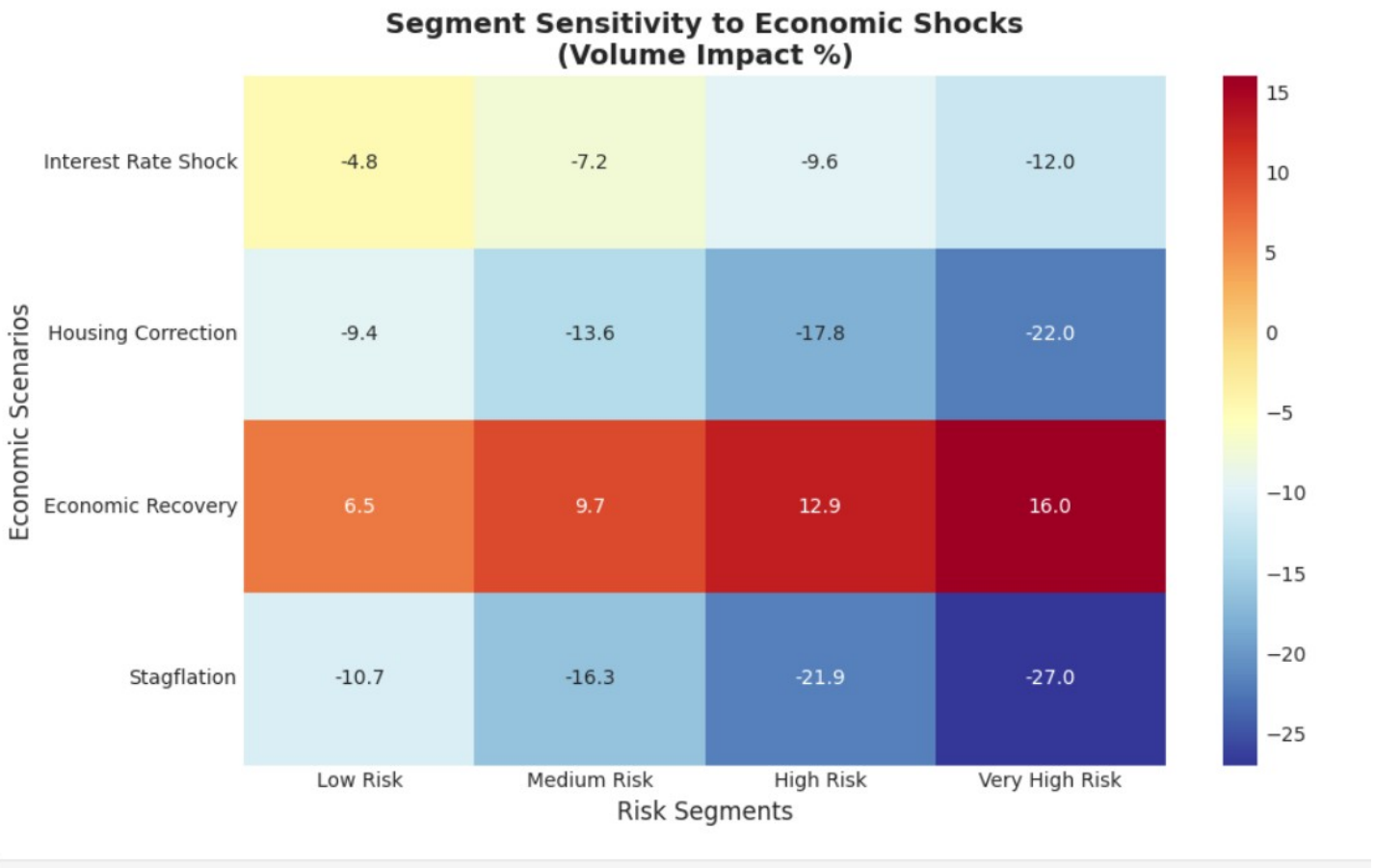


Figure 10: Volume impact across risk segments by economic scenario

Interest rate shock

Negative across all segments, worsening with risk

Housing correction

Significant; most severe for Medium to Very High risk

Economic recovery

Positive; gains increase with risk

Stagflation

Strong negative, especially High and Very High risk

High-risk borrowers fall up to 27% in downturns but recover fastest (up to +16%).

Housing market health assessment

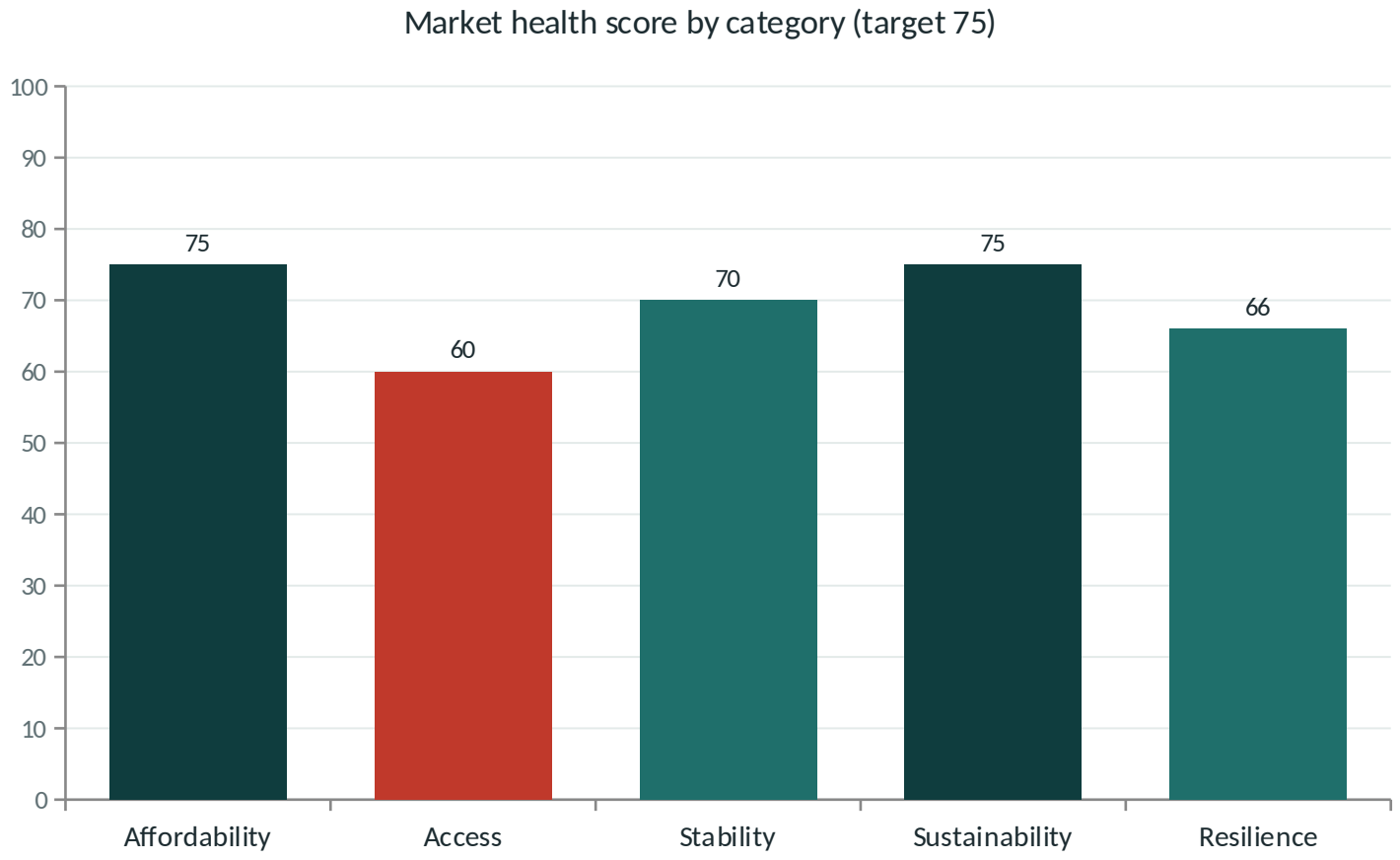


Figure 11: Market health score breakdown by category

OVERALL HEALTH SCORE

69.2 / 100

Moderately healthy: stable, with room to improve

Strongest

Affordability

75

Weakest

Access

60

Policy recommendations

HIGH PRIORITY

Credit access optimization

- Review serviceability buffers
- Implement tiered LVR limits
- Streamline the application process

MEDIUM PRIORITY

Risk-based capital framework

- Introduce risk-weighted capital rules
- Mandatory stress testing for lenders

MEDIUM PRIORITY

Affordability support schemes

- Expand shared equity schemes
- Enhance first home buyer grants and guarantees

Policies compound to lift market health to 97

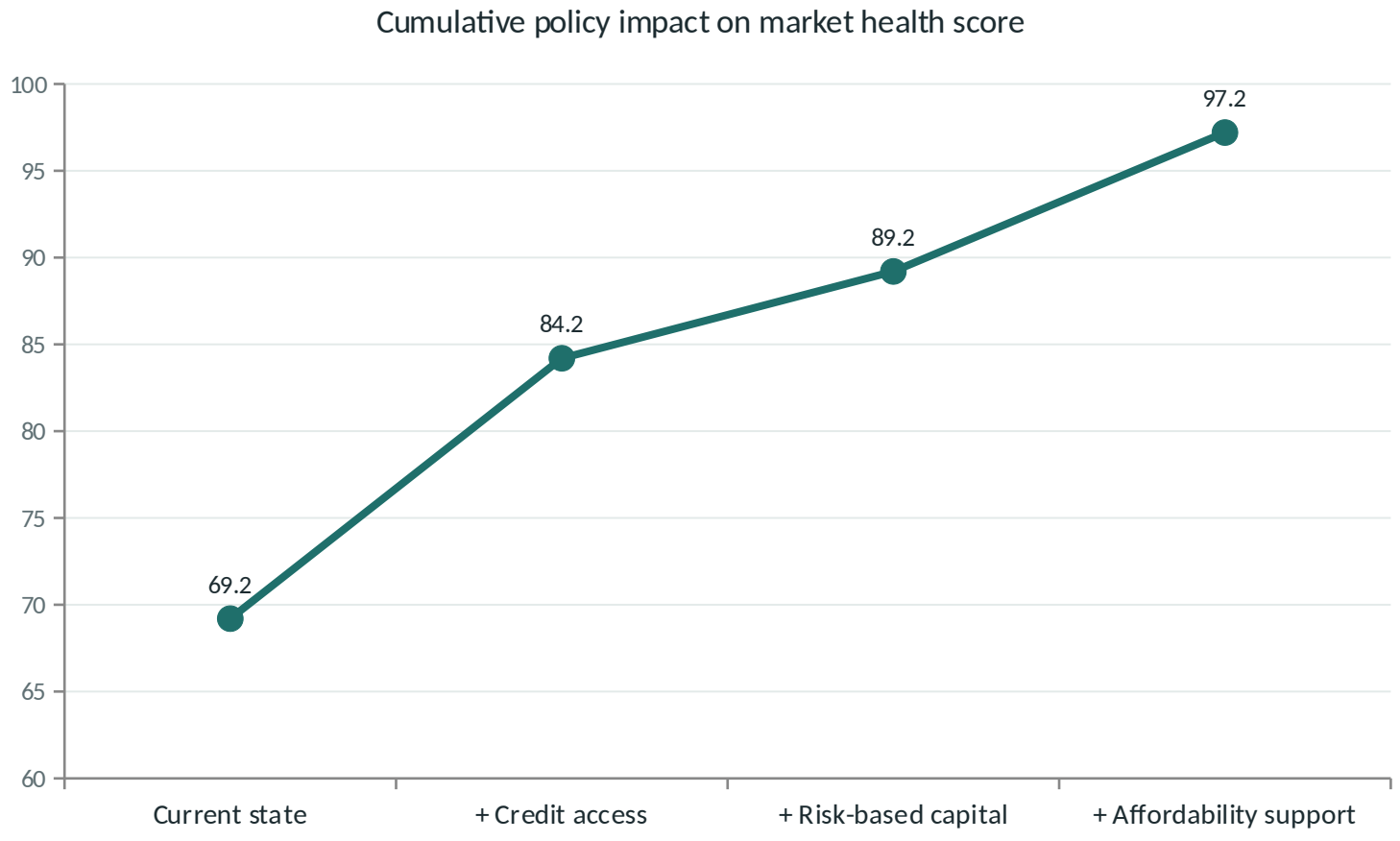


Figure 12: Cumulative policy impact on market health

69.2 Current state

84.2 After credit access optimization (+15.0)

89.2 After risk-based capital framework (+5.0)

97.2 After affordability support (+8.0)

Policies have cumulative, reinforcing effects.

It is getting harder to get a home loan

Tighter lending rules

- Loans are safer
- Fewer people qualify

Weak economy

- Fewer people qualify
- More people can't repay

People who already struggle are hit the hardest.

Policy recommendation

- Make loans easier for creditworthy borrowers
- Keep banks strong
- Support those struggling with deposits
- **Safely increase access and improve market health**

Conclusion & strategic impact

Key deliverables

- Random forest default model: 95% accuracy, ROC-AUC 0.966
- Four borrower risk tiers with tailored strategies
- Regulatory and stress-test impact analysis

Business value

- 67% of borrowers get instant automated decisions
- 29% margin improvement via risk-based pricing
- 20-25% reduction in portfolio default risk
- 55% more efficient capital utilization

Strategic benefits

- Balanced growth: tiered strategy per segment
- Regulatory ready: APRA-aligned, transparent decisions
- Fair lending: risk-appropriate pricing
- Economic resilience: built-in stress testing

Final recommendation: implement the SAS-powered framework for smarter, fairer and more profitable home loan origination.

Limitations & future work

Limitations

- No date fields, so no link to economic or regulatory time series
- Historical / synthetic data may not reflect current market trends
- Random forest is highly accurate but less interpretable
- Fixed recovery and loss rates, not dynamic across scenarios

Future work

- Integrate economic indicators (interest rates, inflation, unemployment)
- Apply explainable AI (SHAP, LIME) for transparent decisions
- Validate and refine with real bank data
- Extend segmentation with behavioural and regional insights

References

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Thank you

Questions and discussion

Jannathul Firdouz | Master of Analytics, AUT



A

Appendix

Calculation details, scoring methodology and supporting analysis

Instant decisions: 67.3%

RESULT

67.3%

of applications auto-approved

METHODOLOGY

Direct count from the model's loan decision distribution.

Inputs

```
Total applications       = 1,789
Auto-approved            = 1,204
```

Calculation

```
Auto-approval rate = (1,204 / 1,789) x 100
                   = 67.3%
```

Result

67.3% instant automated decisions

Portfolio risk reduction: 20-25%

RESULT

23.0%

relative reduction in default rate
(claimed conservatively as 20-25%)

METHODOLOGY

Expected value calculation with conservative assumptions.

Current state

```
Defaults 357 / customers 1,789 = 20.0% default rate
```

After strategy (decline Very High Risk)

```
VHR customers = 182, assumed default rate 60%
Expected VHR defaults = 182 x 0.60 = 109
New defaults = 357 - 109 = 248
New portfolio = 1,789 - 182 = 1,607
New default rate = 248 / 1,607 = 15.4%
```

Risk reduction

```
(20.0% - 15.4%) / 20.0% = 23.0%
```

Profitability improvement: 29.1%

RESULT

29.1%

margin improvement per approved loan

METHODOLOGY

Margin comparison on approved loans. This is margin per approved loan, not total revenue.

Scenario 1: flat pricing (traditional)

Flat interest rate = 7.0%, all 1,789 customers approved

Scenario 2: risk-based pricing

Average risk-based rate = 9.04%

Approved = 1,204 + 248 + 79 = 1,531

Approval rate = 1,531 / 1,789 = 85.6%

Relative margin improvement

(9.04 - 7.0) / 7.0 x 100 = 29.1%

Capital utilization: 55.1%

RESULT

55.1%

reduction in risk-weighted assets

METHODOLOGY

APRA-standard risk-weighted assets (RWA) calculation.

Worst case: all loans at highest risk weight

```
Total loans = $33,504,614
Worst-case RWA = $33,504,614 x 1.50 = $50,256,921
```

Actual: risk-weighted by segment

```
Low (x0.20) $4,193,423  |  Medium (x0.50) $11,255,250
High (x1.00) $6,210,150  |  Very High (x1.50) $922,575
Total actual RWA = $22,581,897
```

Efficiency improvement

```
Savings = $27,675,024
$27,675,024 / $50,256,921 = 55.1%
```

Housing market health: category scoring

Category (weight)	What it measures	Scoring rule	Target
Affordability (25%)	Average interest rate for the low-risk segment	≤6.5%: 85 | ≤7.5%: 75 | ≤8.5%: 65 | >8.5%: 55	≥75 points
Access (25%)	Origination rate (share of applicants who receive loans)	≥60%: 80 | ≥50%: 70 | ≥40%: 60 | ≥30%: 50 | <30%: 40	≥60% access → ≥75
Stability (20%)	Share of High and Very High risk loans in originated volume	≤5%: 90 | ≤10%: 80 | ≤15%: 70 | ≤20%: 60 | >20%: 50	≤10% → ≥80
Sustainability (15%)	Diversity of active loan segments (volume >10% of total)	Active segments / total segments x 100	≥75% active → ≥80
Resilience (15%)	Worst negative volume impact across economic scenarios	100 - 1.5 x max negative impact (40 if impact ≥ 0)	≥75 points
Overall health	Weighted average of the five category scores	Weights shown in brackets	≥75 points

Table 6: Housing market health metrics and scoring methodology

Supporting metrics for the health score

Metric	Description	Calculation	Target
Access rate	Share of applicants who successfully originated loans	Total originated loans / total applicants	≥60%
High-risk share	Share of originated loans that are High or Very High risk	(High + Very High originated) / total originated x 100	≤10%
Average interest rate	Average rate for the low-risk segment	Mean rate, Low Risk segment	≤7.5%
Active segments	Segments with originated volume above 10% of total	Count of segments with volume >10%	≥75% active
Total segments	Number of loan segments analysed	Count of segments	-

Real-world deployment output

```
1. INDIVIDUAL APPLICANT ASSESSMENT

═══════════════════════════════════════════════════════════
🏠 HOME LOAN ORIGINATION DECISION
═══════════════════════════════════════════════════════════
Applicant ID: APP_1001
Submission Time: 2025-10-17 11:27:30
──────────────────────────────────────────────────
KEY FINANCIAL METRICS:
  • Debt-to-Income Ratio: 35.2%
  • Loan Amount: $250,000
  • Property Value: $350,000
  • Delinquencies: 2
  • Derogatory Marks: 1
  • Credit Inquiries: 2
──────────────────────────────────────────────────
RISK ASSESSMENT:
  • Default Probability: 44.1%
  • Risk Level: MEDIUM RISK
  • Borrower Segment: Medium Risk
──────────────────────────────────────────────────
🟢 FINAL DECISION: APPROVED 🟢
💡 Recommendation: Standard terms with regular monitoring
═══════════════════════════════════════════════════════════
```

In production, this is the clear, actionable output delivered to loan officers within seconds.

Risk probability score

44.1% chance of default

Clear categorisation

Medium Risk segment

Specific recommendation

Standard terms with monitoring

Transparent reasoning

Based on measurable financial metrics

Choosing the number of segments

1.617

Davies-Bouldin index

Good separation between the 4 segments

868.6

Calinski-Harabasz index

Well-defined, distinct segments

0.192

Silhouette score

Acceptable level of cluster separation

K = 4

Multiple validation indices together justify four segments, which also align with the practical, interpretable risk tiers banks use.

Portfolio action matrix

GROW

Low risk, large segment

524 applicants | 9.7% default rate

- Best rates (6.5 to 7.5%)
- Streamlined approval
- Premium marketing

Action: Fast-track and invest

MANAGE

Medium risk, medium segment

323 applicants | 17.7% default rate

- Moderate rates
- Standard checks
- Ongoing support

Action: Maintain and monitor

MONITOR

High risk, large segment

759 applicants | 27.9% default rate

- Higher rates
- More due diligence
- Collateral required

Action: Tighten controls

MITIGATE

Very high risk, small segment

182 applicants | 72.6% default rate

- Mostly declined
- Strict approvals only
- Focus on recoveries

Action: Minimise exposure